

Symmetry in nature and in technology

Where the mathematics of the last two lessons actually appears — in crystals, in living things, and in the things engineers build.

LESSON 12

1 × 40 MIN

GEOMETRY 11, §7

IGCSE E4.6

LESSON CLOCK

8'

12'

12'

10'

3'

Symmetry in living things	8 min
Crystals	12 min
Engineering and architecture	12 min
Uzbek ornament	10 min
Homework	3 min

LEARNING OBJECTIVES

- Identify the symmetry of a natural or manufactured object.
- Explain why a particular symmetry is useful in a design.
- Name the crystal systems by their symmetry.
- Present one object and its symmetry to the class.

TEXTBOOK

National	pp. 65–70
Cambridge	Core & Extended, pp. 363–366
Workbook	Project sheet S1

01

Explanation

Living things

ORGANISM	SYMMETRY	WHY IT HELPS
most animals	bilateral — one plane	a front and a direction of travel
starfish	radial, order 5	no preferred direction; food may come from anywhere
most flowers	radial, order 3–8	visible to a pollinator from any side
snowflake	radial, order 6	the hexagonal geometry of the water molecule
DNA	a helix — screw symmetry	compact storage of a long chain

A USEFUL OBSERVATION

Animals that **move** in a chosen direction are bilaterally symmetric; those that stay put, or drift, are radially symmetric. Motion selects a plane; stillness does not.

Crystals

A crystal is a lattice — one **unit cell** repeated by translation in three directions. The seven crystal systems are classified purely by their symmetry:

SYSTEM	CELL	EXAMPLE
cubic	$a = b = c$, all angles 90°	salt, diamond, pyrite
tetragonal	$a = b \neq c$, all 90°	zircon
orthorhombic	$a \neq b \neq c$, all 90°	sulfur
hexagonal	a hexagonal prism cell	quartz, ice

A cubic cell carries all 9 planes and 13 axes of the cube — which is why salt cleaves into little cubes, and why a diamond can be cut along predictable planes.

Engineering, architecture, ornament

- **Wheels, gears, turbines** — radial symmetry of order n balances the rotating mass; an unbalanced wheel destroys its bearing.
- **Bridges and domes** — a plane of symmetry halves the load calculation and makes the structure predictable under wind from either side.
- **Aircraft** — bilateral symmetry keeps the centre of mass on the centre line.
- **Screw threads** — helical symmetry; the handedness matters, which is why a left-handed bolt is a real and separate object.

Uzbek ornament. The *girih* patterns of Samarkand and Bukhara are built from a small set of tiles with 5- and 10-fold rotational symmetry, repeated by translation. The tilework of the Registan is a mathematical object: a plane group, chosen and executed by hand centuries before the classification of such groups was written down.

THE PROJECT

Each learner brings one object — natural or made — photographs or sketches it, and states its symmetry precisely: how many planes, how many axes and of what order, and whether it has a centre. Two present next lesson.

02 Worked examples**EXAMPLE 1** State the symmetry of a starfish.

- ① Five identical arms.
Rotational order 5.
- ② One mirror line through each arm.
5 planes in three dimensions.
- ③ No centre — the top and bottom differ.

ANSWER

Radial, order 5, with 5 planes; no centre

EXAMPLE 2 Why must a turbine blade assembly be rotationally symmetric?

- ① An asymmetric mass has a centre of mass off the axis.
- ② That produces a rotating force on the bearings.
- ③ At high speed the force is destructive.

ANSWER

To keep the centre of mass on the rotation axis and avoid vibration

EXAMPLE 3 A salt crystal is cubic. How many planes of symmetry has one grain?

- ① A cube has 3 face planes.
- ② And 6 through opposite edges.

ANSWER

9

03 Interactive model

Bring a real object — a nut, a flower, a snowflake photograph — and count its symmetries aloud.

Symmetry around us

Most animals have:

radial symmetry

bilateral symmetry

no symmetry

a centre of symmetry

A snowflake has rotational order:

4

5

6

8

A cubic crystal has how many planes?

3

6

9

12

A screw thread has:

plane symmetry

helical symmetry

central symmetry

no symmetry

Girih patterns commonly show rotational order:

3 and 4

5 and 10

7

12 only

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Bilateral symmetry	Ikki tomonlama simmetriya	Двусторонняя симметрия
Radial symmetry	Radial simmetriya	Радиальная симметрия
Crystal lattice	Kristall panjara	Кристаллическая решётка
Unit cell	Elementar yacheyka	Элементарная ячейка
Helix	Spiral	Спираль
Tessellation	Parketlash	Паркет (замощение)
Balance (in design)	Muvozanat	Равновесие
Girih pattern	Girih naqshi	Гирих
Chirality	Xirallik	Хиральность

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Radial symmetry suits organisms that:

move fast forward

stay in one place

fly

burrow

A unit cell is:

a whole crystal

the smallest repeating block

a plane of symmetry

an axis

A left-handed and a right-handed screw are:

identical

mirror images

rotations of each other

the same object

Symmetry in a bridge mainly helps with:

appearance only

load calculation and predictability

cost of paint

nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Symmetry of a starfish

ANSWER radial, order 5

2. Symmetry of a human body (externally)

ANSWER bilateral, one plane

3. Rotational order of a snowflake

ANSWER 6

4. Symmetry of a wheel with 8 spokes

ANSWER radial, order 8

5. Planes of a cubic salt crystal

ANSWER 9

6. Symmetry of DNA

ANSWER helical

7. Symmetry of a flower with 5 petals

ANSWER radial, order 5

Medium · the standard question

7 PROBLEMS

1. Why are aircraft bilaterally symmetric?

ANSWER To keep the centre of mass on the centre line

2. Why must a turbine be radially symmetric?

ANSWER To avoid vibration from an off-axis mass

3. Name two cubic minerals

ANSWER salt, pyrite (also diamond)

4. Which crystal system has $a = b \neq c$ with right angles?

ANSWER tetragonal

5. What rotational orders appear in girih patterns?

ANSWER 5 and 10

6. Why do most flowers have radial symmetry?

ANSWER Pollinators approach from any direction

7. Give a manufactured object with exactly one plane of symmetry

ANSWER e.g. a chair, a car exterior

Hard · stretch and reason

7 PROBLEMS

1. Explain why 5-fold symmetry cannot tile the plane periodically

ANSWER Pentagons leave gaps; only orders 2, 3, 4, 6 tile

2. How do girih tilings achieve 5-fold local symmetry then?

ANSWER By combining several tile shapes — a quasi-periodic arrangement

3. A crystal has 3 planes and a centre. Which system?

ANSWER orthorhombic

4. Why is chirality important in chemistry?

ANSWER Mirror-image molecules can behave completely differently in the body

5. Design a logo with rotational order 3 and no plane of symmetry

ANSWER e.g. a three-bladed spiral

6. Count the symmetries of a standard hexagonal pencil

ANSWER 7 planes, 7 axes

7. Why does a soap bubble take a spherical shape?

ANSWER Maximum symmetry minimises surface area for a fixed volume

07 Homework

Homework — 4 tasks

Bring your object, or a clear photograph of it, to the next lesson.

1. Choose one object — natural or manufactured — and state its symmetry precisely: planes, axes with orders, and whether it has a centre.
2. Photograph or sketch a piece of Uzbek tilework and identify its rotational order.
3. Explain in three sentences why moving animals are bilaterally symmetric and stationary ones are radially symmetric.
4. Name the four crystal systems in the lesson with one example each.